

MODELS	METHODS	GSM8K	AQUA	ECQA	AVERAGE (Δ)
LLAMA-7B	VFT	36.48 $\pm$ 0.92	31.19 $\pm$ 0.28	70.40 $\pm$ 1.07	46.02 (-)
	RFT	39.75 $\pm$ 1.03	32.81 $\pm$ 1.48	72.23$\pm$0.11	48.28 ($\uparrow$ 2.26)
	AFT ($\mathcal{L}_A^{DC}$)	40.43$\pm$1.04	33.01 $\pm$ 0.95	72.23$\pm$0.43	48.55 ($\uparrow$ 2.53)
	AFT ($\mathcal{L}_A^{BC}$)	40.26 $\pm$ 0.36	33.20$\pm$1.24	72.15 $\pm$ 0.57	48.53 ($\uparrow$ 2.51)
LLAMA2-7B	VFT	40.71 $\pm$ 0.16	31.49 $\pm$ 1.96	72.34 $\pm$ 0.22	48.18 (-)
	RFT	43.65 $\pm$ 0.13	33.25 $\pm$ 1.23	73.86$\pm$0.38	50.25 ($\uparrow$ 2.07)
	AFT ($\mathcal{L}_A^{DC}$)	44.25$\pm$0.43	33.49$\pm$0.63	73.71 $\pm$ 0.65	50.75 ($\uparrow$ 2.57)
	AFT ($\mathcal{L}_A^{BC}$)	44.16 $\pm$ 0.81	32.89 $\pm$ 0.98	73.23 $\pm$ 0.82	50.09 ($\uparrow$ 1.91)
LLAMA-13B	VFT	42.07 $\pm$ 0.15	33.91 $\pm$ 0.60	72.74 $\pm$ 0.43	49.57 (-)
	RFT	46.13 $\pm$ 1.41	34.29 $\pm$ 1.28	75.03$\pm$0.35	51.80 ($\uparrow$ 2.23)
	AFT ($\mathcal{L}_A^{DC}$)	46.31 $\pm$ 1.52	34.49 $\pm$ 1.21	74.32 $\pm$ 0.09	51.70 ($\uparrow$ 2.13)
	AFT ($\mathcal{L}_A^{BC}$)	46.46$\pm$0.28	34.79$\pm$0.37	74.53 $\pm$ 0.68	51.93 ($\uparrow$ 2.36)
LLAMA2-13B	VFT	47.29 $\pm$ 1.24	34.68 $\pm$ 1.36	74.76 $\pm$ 0.56	52.24 (-)
	RFT	50.12 $\pm$ 1.57	34.95 $\pm$ 0.88	76.21 $\pm$ 0.80	53.75 ($\uparrow$ 1.51)
	AFT ($\mathcal{L}_A^{DC}$)	50.67 $\pm$ 1.16	35.78$\pm$0.45	76.42 $\pm$ 0.82	54.29 ($\uparrow$ 2.05)
	AFT ($\mathcal{L}_A^{BC}$)	51.03$\pm$0.54	35.49 $\pm$ 1.19	76.57$\pm$0.83	54.36 ($\uparrow$ 2.12)

Table 2: The accuracy of different methods on three reasoning datasets. Δ denotes the improvement compared to VFT. AFT significantly outperforms VFT, and is slightly better than RFT (Yuan et al., 2023a). Note that RFT is a concurrent work to ours.

new augmented training data to train LLMs, which is proven to be a very strong baseline; 3) **RRHF**: Rank Responses to align Human Feedback (RRHF) (Yuan et al., 2023b), which takes candidate ranking into account and distinguishes different candidates through a pair-wise ranking loss; 4) **PRO**: Preference Ranking Optimization (PRO) (Song et al., 2023a), which takes candidate ranking into account and distinguishes different candidates through a ranking loss with a dynamic temperature.

Metrics We use the accuracy to measure the model performance. Specifically, we conduct 3 runs with 3 different seeds and report the average results with the standard deviation.

5.2 RESULTS WITH BINARY FEEDBACK

Table 2 displays the results of different fine-tuning methods on three reasoning datasets. As is shown: **1)** AFT significantly outperforms VFT on all three datasets, improving the average accuracy by 1.91% $\sim$ 2.57% for all models, showing the effectiveness of AFT; **2)** Our concurrent work RFT also expresses notable improvement compared with VFT. However, the original RFT paper only treats RFT as a simple data augmentation method without explaining the reasons behind its notable improvement. Our alignment perspective can provide an explanation for the effectiveness of RFT, i.e., RFT can alternatively be regarded as an alignment strategy that bolsters the scores of numerous positive COTs and thus can alleviate the assessment misalignment problem of VFT; **3)** Our proposed two constraint alignment strategies slightly outperform RFT with the binary feedback. In addition, our AFT can be also easily extended to utilize the ranking feedback that RFT can not well utilize. These results demonstrate the importance of revealing the assessment misalignment problem of VFT and the effectiveness of our AFT approach.

5.3 RESULTS WITH RANKING FEEDBACK

As described in Section 4.4, our AFT can also be easily adapted to the ranking situation where we can obtain the quality ranking sequence of generated COTs. Table 3 illustrates the results of different methods in the GSM8k-RANK. As is shown: **1)** Our AFT surpasses all other methods, demonstrating its effectiveness with ranking feedback. For instance, AFT exceeds the strongest baseline RFT by 0.88% in average accuracy. This superiority can be attributed to AFT’s ability to help LLMs recognize quality differences among any given pair in a ranking context, while RFT only focuses exclusively on optimizing the probability of the highest-quality examples; **2)** Prior methods utilizing ranking loss

METHODS	LLAMA-7B	LLAMA-13B	LLAMA2-7B	LLAMA2-13B	AVERAGE (Δ)
VFT	20.82 $\pm$ 0.71	24.12 $\pm$ 0.42	24.08 $\pm$ 0.22	30.28 $\pm$ 1.46	24.83 (-)
RFT	25.09 $\pm$ 1.18	28.21 $\pm$ 0.86	28.25 $\pm$ 0.78	34.53 $\pm$ 0.51	29.02 ($\uparrow$ 4.19)
RRHF	7.51 $\pm$ 0.56	9.92 $\pm$ 0.82	9.21 $\pm$ 0.25	13.35 $\pm$ 1.26	10.00 ($\downarrow$ 14.8)
PRO	18.73 $\pm$ 0.31	20.34 $\pm$ 1.51	21.40 $\pm$ 0.92	23.55 $\pm$ 0.98	21.00 ($\downarrow$ 3.82)
AFT (L_A^{RBC})	26.08$\pm$1.05	28.97$\pm$0.35	29.05$\pm$0.75	35.48$\pm$1.35	29.90 ($\uparrow$ 5.07)

Table 3: Test accuracy of different methods on GSM8K trained with GSM8K-RANK.

METHODS	WITHOUT CONSTRAINT			WITH CONSTRAINT (OURS)		
	$\mathbf{T_{Accuracy}}$	$\mathbf{A_{Accuracy}}$	$\mathbf{PPL (\downarrow)}$	$\mathbf{T_{Accuracy}}$	$\mathbf{A_{Accuracy}}$	$\mathbf{PPL (\downarrow)}$
VFT	20.82$\pm$0.71	68.72 $\pm$ 1.48	1.60$\pm$0.01	20.82 $\pm$ 0.71	68.72 $\pm$ 1.48	1.60 $\pm$ 0.01
RRHF	7.51 $\pm$ 0.56	87.44 $\pm$ 1.28	1.80 $\pm$ 0.01	25.53 $\pm$ 0.27	79.89 $\pm$ 0.60	1.35$\pm$0.01
PRO	18.73 $\pm$ 0.31	86.58 $\pm$ 1.09	2.34 $\pm$ 0.02	25.82 $\pm$ 0.48	80.34 $\pm$ 0.97	1.45 $\pm$ 0.01
AFT (L_A^{RBC})	7.03 $\pm$ 0.98	88.89$\pm$0.78	7.81 $\pm$ 0.03	26.08$\pm$1.05	81.36$\pm$0.78	1.37 $\pm$ 0.01

Table 4: Task accuracy ($\mathbf{T_{Accuracy}}$) and assessment accuracy ($\mathbf{A_{Accuracy}}$) on GSM8K for LLama-7B, which is fine-tuned by different methods (with or without constraint) on GSM8K-Rank. $\mathbf{PPL (\downarrow)}$, lower is better) denotes the average perplexity of all positive COTs.

have a substantial negative impact on model performance. For example, integrating RRHF loss into VFT leads to a 14.8% reduction in accuracy. In fact, the performance reduction is also observed in their own paper (Song et al., 2023a), which demonstrates that ranking loss often enhances the reward of LLMs, yet results in lower BLEU scores. However, they do not identify the cause, and in this paper, we find that a potential reason for the performance decline is the absence of a constraint in their loss, which we will discuss in Section 6.1.

6 ANALYSIS

6.1 DELVE DEEPLY INTO RECENT RANKING LOSSES FOR ALIGNMENT

Our experiments on GSM8K-RANK show that adding ranking loss will harm the model performance. We think the reason is that previous alignment ranking losses will unreasonably decrease the score of non-optimal COTs (Please refer to Appendix D for our detailed analysis). To empirically validate this hypothesis, we add a detached constraint to these two ranking losses similar to $\mathcal{L}_A^{RDC1}$ (Equation 12). Consequently, these ranking losses will only make the scores of higher-quality COTs larger than those of lower-quality ones, without explicitly decreasing the scores of COTs with lower quality. Table 4 illustrates the final accuracy $\mathbf{T_{Accuracy}}$ of different methods in the testing set, the assessment accuracy $\mathbf{A_{Accuracy}}$ and average perplexity of positive COTs $\mathbf{PPL}$ in the training set¹. As is shown: **1)** Without the constraint strategy, all three ranking losses harm the model performance, leading to higher perplexity and lower final task accuracy compared to VFT; **2)** We observe that the task accuracy of PRO does not decline as significantly as RRHF and AFT. We think this is because PRO employs a dynamic temperature that reduces the negative score in a more reasonable manner (Please refer to Appendix D.3 for details); **3)** By adding the constraint, all ranking losses can not only improve two accuracies but also decrease the perplexity. These results show the importance of constraint for other ranking losses for alignment. Furthermore, we also conduct a case study in Appendix E to intuitively show the model degradation without constraint.

6.2 EFFECTIVENESS OF THE NUMBER OF CANDIDATE COTs

As described in Section 4.1, AFT samples k candidate generation results to align LLMs. In this section, we explore the influence of k . We sampled 0, 8, 16, 32, and 64 results from the VFT-LLama-7B, and then de-duplicated these sampling results. Then, we train LLama-7B on the de-duplicated datasets.

¹For each question in the training set, we sample new COTs (three positive and three negative COTs, respectively) that are different from training COTs for evaluation.

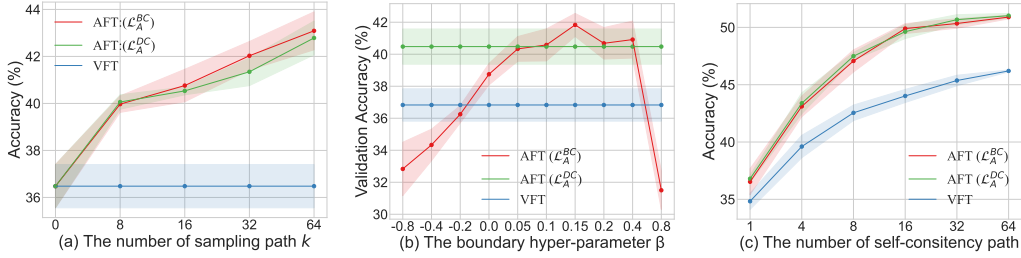


Figure 2: Variation of accuracy with (a): different number of sampling COTs for training; (b) different boundary constraint hyper-parameter; (c) different number of voting paths of self-consistency.

METHODS	GSM8K	AQUA	ECQA	MMLU	AVERAGE (Δ)
VFT	35.72 ± 0.95	32.95 ± 0.98	69.25 ± 0.74	37.52 ± 1.03	43.86 (-)
AFT ($\mathcal{L}_A^{DC}$)	40.24 ± 0.63	33.72 ± 0.92	71.38 ± 0.64	39.25 ± 0.35	46.15 ($\uparrow$ 2.29)
AFT ($\mathcal{L}_A^{BC}$)	40.00 ± 0.69	33.45 ± 0.56	71.48 ± 0.89	38.89 ± 0.70	45.96 ($\uparrow$ 2.10)

Table 5: Comparison of VFT- and AFT-LLama-7B with training data ‘‘GSM8K+AQUA+ECQA’’ on three in-domain benchmarks and an out-of-domain benchmark MMLU.

As shown in Figure 2(a), we can see that AFT can consistently improve the model performance with k improvement, which is a promising result. We think the reason is that with large k , the AFT will have more data to help the LLM perceive the quality of different COT paths, which enhances the final performance. This growing accuracy shows the effectiveness and the potential of AFT.

6.3 ABLATION ON THE BOUNDARY VALUE

The boundary constraint term of AFT requires a hyper-parameter β to regulate the boundary. In this section, we conduct an ablation study to demonstrate the impact of varying β values. As depicted in Figure 2(b), the performance initially increases and subsequently decreases as β ranges from -0.8 to 0.8. These findings align with expectations, as a small β cannot effectively widen the score gap between high-quality and low-quality COTs, while an overly large β may result in excessively low scores for non-optimal COTs, thereby compromising the model’s generative abilities. In conclusion, the results emphasize the importance of the boundary constraint term and indicate that the value of β can significantly affect model performance. Therefore, it is essential to carefully adjust this value when using our boundary constraint alignment loss.

6.4 EFFECTIVENESS OF AFT WITH SELF-CONSISTENCY

Self-consistency is a highly effective strategy for improving LLM’s reasoning performance. This method involves sampling multiple COTs and utilizing a voting process to determine the final answer during inference. AFT samples COTs for training to develop better LLMs. Both methods utilize COTs to enhance the model’s reasoning ability. In this section, we explore the combination of AFT and Self-Consistency. As illustrated in Figure 2(c), as the number of paths increases, the improvement of AFT is more significant than VFT, demonstrating that AFT effectively enhances self-consistency. We believe the reason is that AFT helps models learn to assess the quality of different COTs by encouraging larger scores for high-quality COTs compared to low-quality ones. This means that high-quality COTs are more likely to be sampled, and thus, AFT can enhance self-consistency.

6.5 EFFECTIVENESS OF AFT ON THE MULTI-TASK AND OUT-OF-DOMAIN SITUATIONS

To further demonstrate the effectiveness and versatility of AFT, we investigate its performance in multi-task scenarios. We combine the training sets of three datasets and use both AFT and VFT to train the LLama-7B model. As depicted in Table 5, AFT is able to simultaneously enhance the performance of all corresponding test sets. Additionally, we evaluate both AFT and VFT on the MMLU (zero-shot), an out-of-distribution benchmark, and AFT also outperforms VFT. These results